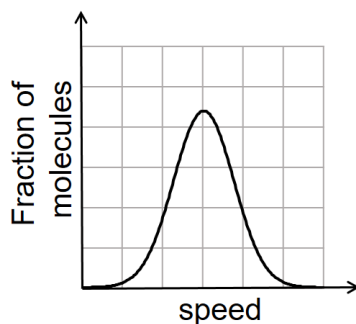


PART II: CHEMISTRY

SECTION-1 (Maximum Marks: 18)

- This section contains SIX (06) questions
- Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following making scheme:
 Full Marks : +3 If **ONLY** the correct option is chosen;
 Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
 Negative Marks : -1 In all other cases

- *1. If the distribution of molecular speeds of a gas is as per the figure shown below, then the ratio of the most probable, the average, and the root mean square speeds, respectively, is



- (A) 1:1:1
 (B) 1 : 1 : 1.224
 (C) 1 : 1.128 : 1.224
 (D) 1 : 1.128 : 1

Sol.

B

Since the graph is symmetrical about the peak. So, the average speed and most probable speed will coincide

but rms speed will remain $\sqrt{\frac{3RT}{M}}$

$$\text{So, } u_{mp} = \sqrt{\frac{2RT}{M}} = u_{av}$$

$$u_{rms} = \sqrt{\frac{3RT}{M}}$$

$$u_{mp} : u_{av} : u_{rms} = \sqrt{2} : \sqrt{2} : \sqrt{3} = 1 : 1 : \sqrt{\frac{3}{2}}$$

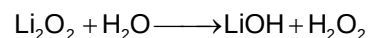
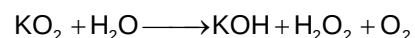
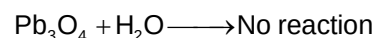
$$= 1 : 1 : 1.224$$

- *2. Which of the following liberates O_2 upon hydrolysis?

- (A) Pb_3O_4 (B) KO_2
 (C) Na_2O_2 (D) Li_2O_2

Sol.

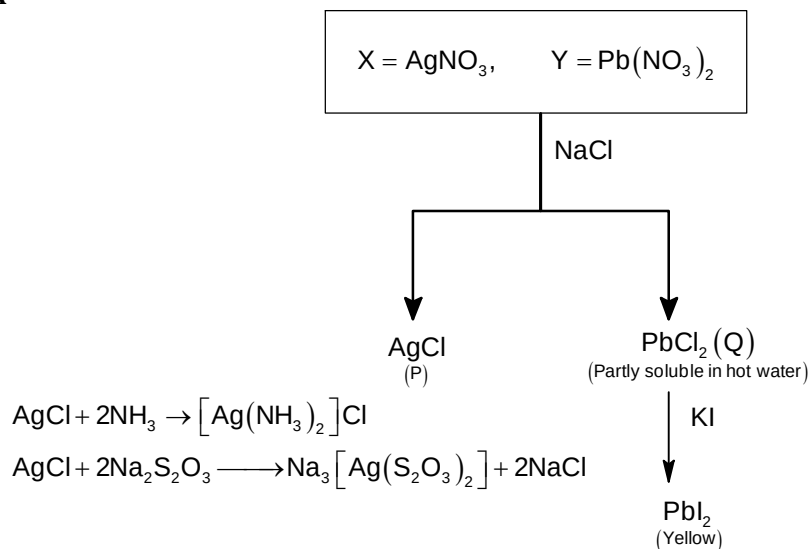
B



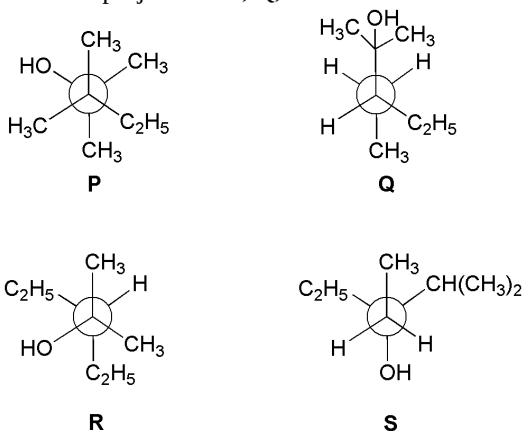
3. A colorless aqueous solution contains nitrates of two metals, **X** and **Y**. When it was added to an aqueous solution of NaCl, a white precipitate was formed. This precipitate was found to be partly soluble in hot water to give a residue **P** and a solution **Q**. The residue **P** was soluble in aqueous NH_3 and also in excess sodium thiosulfate. The hot solution **Q** gave a yellow precipitate with KI. The metals **X** and **Y**, respectively are

- (A) Ag and Pb (B) Ag and Cd
(C) Cd and Pb (D) Cd and Zn

Sol. A



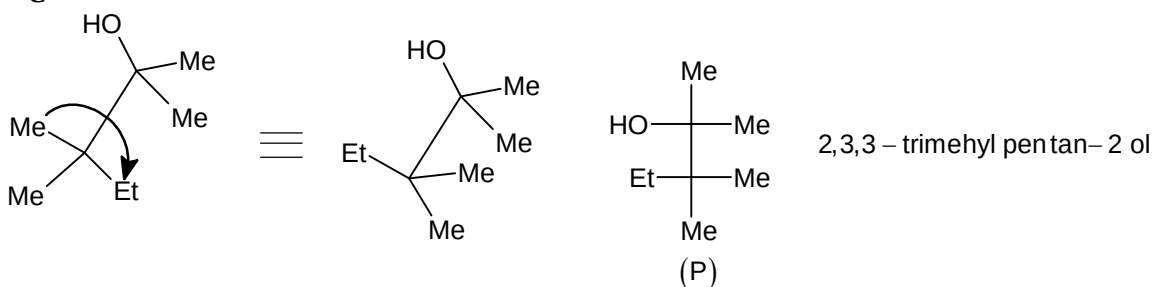
- *4. Newman projections P, Q, R and S are shown below:



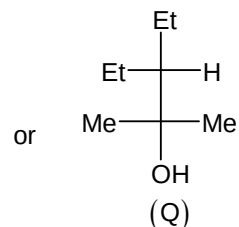
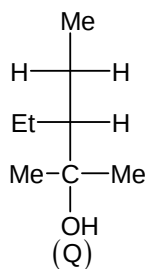
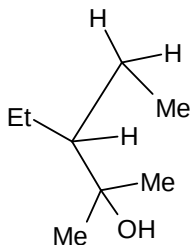
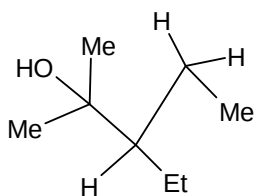
Which one of the following options represents identical molecules?

- (A) P and Q (B) Q and S
(C) Q and R (D) R and S

Sol. C
(I)

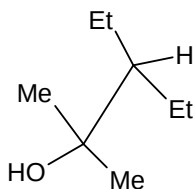


(II)

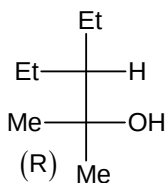
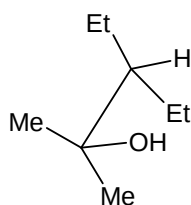


3-ethyl, 2-methyl pentan-2-ol

(III)

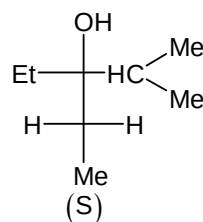
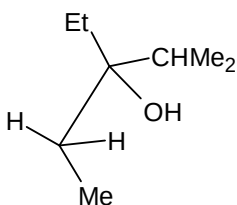
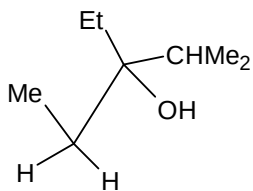


≡



3-ethyl, 2-methylpentan-2-ol

(IV)

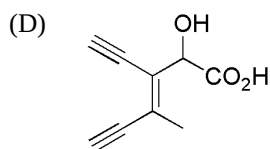
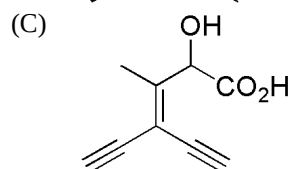
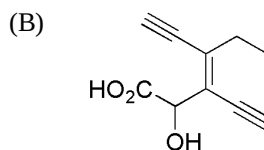
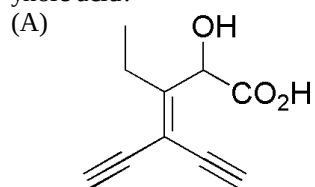
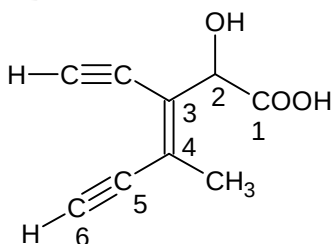


3-ethyl, 2-methyl pentan-3-ol

Only (Q & R) are same.

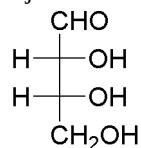
*5.

Which one of the following structures has the IUPAC name 3-ethynyl-2-hydroxy-4-methylhex-3-en-5-ynoic acid?

**Sol. D**

3-ethynyl, 2-hydroxy, 4-methyl hex-3-en-5-ynoic acid

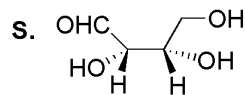
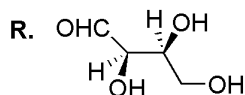
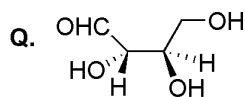
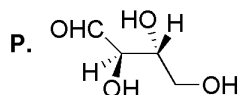
*6. The Fischer projection of D-erythrose is shown below:



D-Erythrose

D—Erythrose and its isomers are listed as P, Q, R and S in **Column-I**. Choose the correct relationship of P, Q, R and S with D-erythrose from **Column-II**.

Column-I



Column-II

1. Diastereomer
2. Identical
3. Enantiomer

(A) P → 2, Q → 3, R → 2, S → 2

(C) P → 2, Q → 1, R → 1, S → 3

(B) P → 3, Q → 1, R → 1, S → 2

(D) P → 2, Q → 3, R → 3, S → 1

Sol.

C

Given structure : (2R, 3R)

P ⇒ (2R, 3R), so, identical. Hence P → 2

Q ⇒ (2S, 3R), so, Diastereomers. Hence Q → 1

R ⇒ (2R, 3S), so, Diastereomers. Hence R → 1

S ⇒ (2S, 3S), so, Enantiomers. Hence S → 3

SECTION-2 (Maximum Marks: 24)

- This section contains **SIX (06)** questions
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is/are correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following making scheme:

Full Marks	: +4	If Only (all) the correct option(s) is/are chosen;
Partial Marks	: +3	If all the four option are correct but ONLY three options are chosen;
Partial Marks	: +2	If three or more option are correct but ONLY two options are chosen, both of which are correct
Partial Marks	: +1	If two or more option are correct but ONLY one option are chosen, and it is a correct option
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	: -2	In all other cases

- *7. In thermodynamics, the P – V work done is given by

$$w = -\int dVP_{\text{ext}}$$

For a system undergoing a particular process, the work done is

$$w = -\int dV \left(\frac{RT}{V-b} - \frac{a}{V^2} \right)$$

This equation is applicable to a

- (A) system that satisfies the van der Waal's equation of state
- (B) process that is reversible and isothermal
- (C) process that is reversible and adiabatic
- (D) process that is irreversible and at constant pressure

Sol. A, B, C

$$W_{\text{rev}} = -\int PdV$$

For one mole of a real gas,

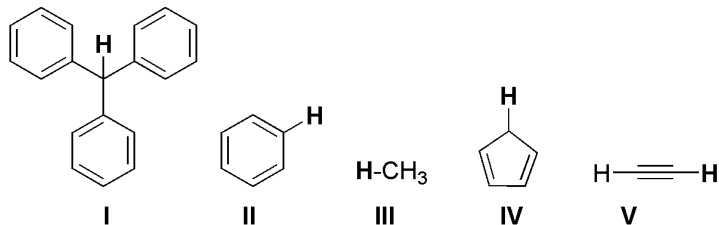
$$\left(P + \frac{a}{V^2} \right) (V - b) = RT$$

$$P = \left(\frac{RT}{(V-b)} - \frac{a}{V^2} \right)$$

$$\text{So, } W_{\text{rev}} = -\int \left(\frac{RT}{(V-b)} - \frac{a}{V^2} \right) dV \dots (1)$$

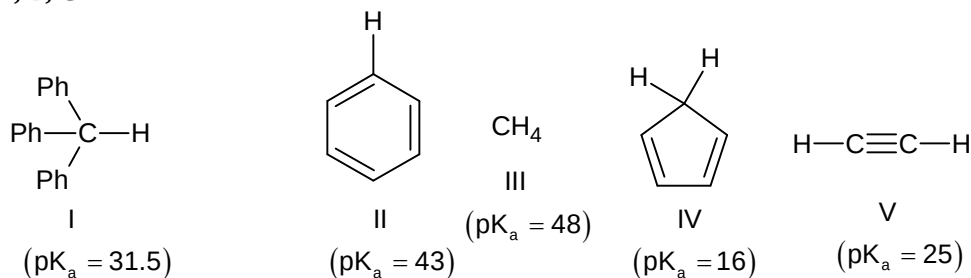
So, above Equation is valid for all reversible processes and for real gases.

- *8. With respect to the compounds **I – V**, choose the correct statement(s).



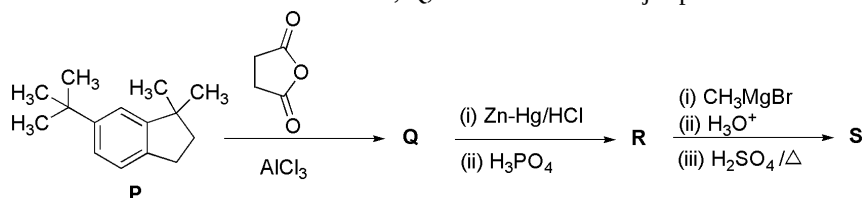
- (A) The acidity of compound I is due to delocalization in the conjugate base
- (B) The conjugate base of compound IV is aromatic
- (C) Compound II becomes more acidic, when it has a $-\text{NO}_2$ substituent
- (D) The acidity of compounds follows the order $\text{I} > \text{IV} > \text{V} > \text{II} > \text{III}$

Sol. A, B, C



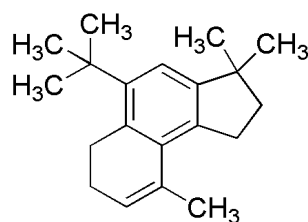
Correct order and acidity = $\text{IV} > \text{V} > \text{I} > \text{II} > \text{III}$ (Decreasing acidity)

*9. In the reaction scheme shown below, Q, R and S are the major products.

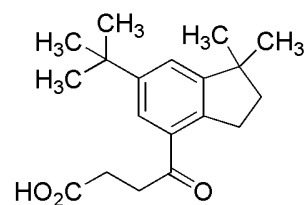


The correct structure of

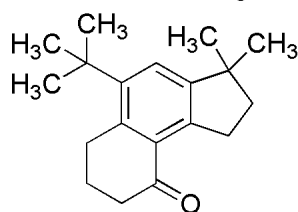
(A) S is



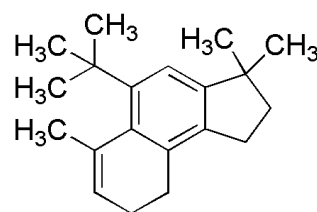
(B) Q is



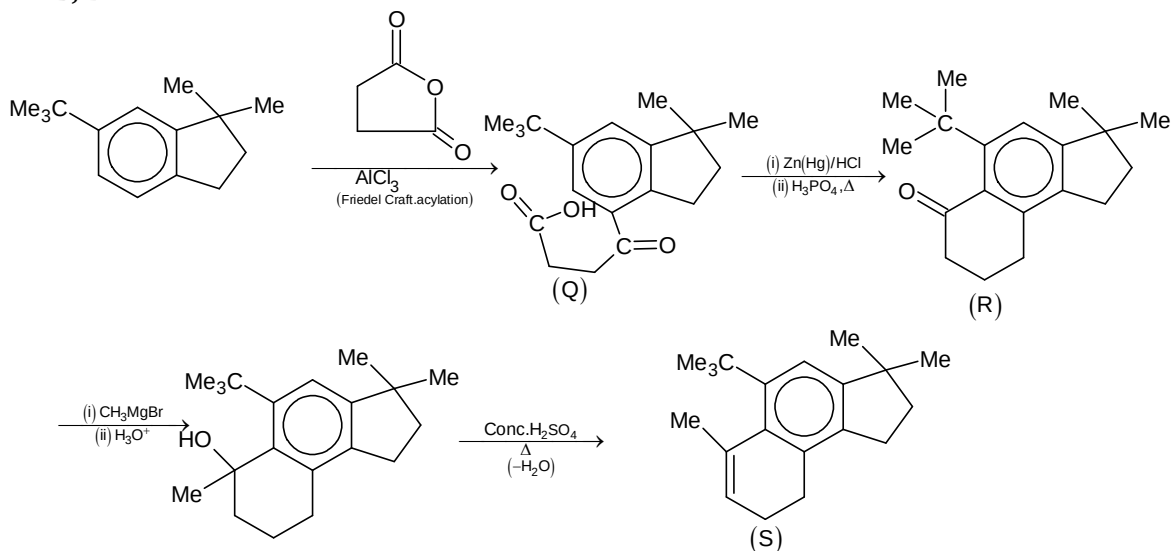
(C) R is



(D) S is



Sol. B, D

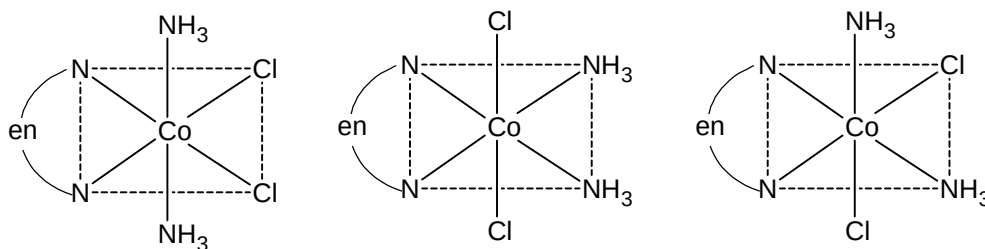


10. Choose the correct statement(s) among the following:
- (A) $[\text{FeCl}_4]^-$ has tetrahedral geometry
- (B) $[\text{Co}(\text{en})(\text{NH}_3)_2\text{Cl}_2]^+$ has 2 geometrical isomers
- (C) $[\text{FeCl}_4]^-$ has higher spin – only magnetic moment than $[\text{Co}(\text{en})(\text{NH}_3)_2\text{Cl}_2]^+$
- (D) The cobalt ion in $[\text{Co}(\text{en})(\text{NH}_3)_2\text{Cl}_2]^+$ has sp^3d^2 hybridization

Sol. A, C

(A) $[\text{FeCl}_4]^-$: Fe^{3+} , weak ligand, so tetrahedral.

(B) $[\text{Co}(\text{en})(\text{NH}_3)_2\text{Cl}_2]^+$: Three geometrical isomers.



(C) $[\text{FeCl}_4]^-$: $N = 5$. $\mu = \sqrt{5(5+2)} = \sqrt{35} \text{ BM}$

$[\text{Co}(\text{en})(\text{NH}_3)_2\text{Cl}_2]^+$: $\text{Co}^{3+} (d^6)$, diamagnetic

(D) $[\text{Co}(\text{en})(\text{NH}_3)_2\text{Cl}_2]^+$: $d^2\text{sp}^3$, $N = 0$, $\mu = 0$ & $[\text{FeCl}_4]^- \Rightarrow \text{Tetrahedral} (sp^3)$

11. With respect to hypochlorite, chlorate and perchlorate ions, choose correct statement(s).
- (A) The hypochlorite ion is strongest conjugate base
- (B) The molecular shape of only chlorate ion is influenced by the lone pair of electrons of Cl
- (C) The hypochlorite and chlorate ions disproportionate to give rise to identical set of ions
- (D) The hypochlorite ion oxidizes the sulfite ion

Sol. A, B, D

(A) $\text{HClO} < \text{HClO}_3 < \text{HClO}_4$ (increasing acidic character)

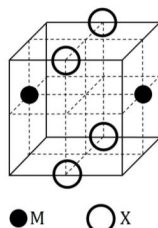
So, ClO^- is the strongest conjugate base.

(B) Only $\text{Cl}-\text{O}^-$ and ClO_3^- has lone pair of electrons on Cl-atom. But $\text{Cl}-\text{O}^-$ is always linear ion as it is diatomic, so, no effect of lone pair on shape but in ClO_3^- due the lone pair on Cl shape becomes pyramidal.

(C) $\text{ClO}^- \longrightarrow \text{Cl}^- + \text{ClO}_3^-$
 $\text{ClO}_3^- \longrightarrow \text{Cl}^- + \text{ClO}_4^-$ (No same set of ions)

(D) $\text{ClO}^- + \text{SO}_3^{2-} \longrightarrow \text{Cl}^- + \text{SO}_4^{2-}$

12. The cubic unit cell structure of a compound containing cation M and anion X is shown below. When compared to the anion, the cation has smaller ionic radius. Choose the correct statement(s).



- (A) The empirical formula of the compound is MX
 (B) The cation M and anion X have different coordination geometries
 (C) The ratio of M – X bond length to the cubic unit cell edge length is 0.866
 (D) The ratio of the ionic radii of cation M to anion X is 0.414

Sol. A, C

It can be assumed to be CsCl type solid (bcc)

$$\text{So, } r_+ + r_- = \frac{\sqrt{3}a}{2} = 0.866$$

Formula of solid = MX

$$0.732 \leq \frac{r_+}{r_-} \leq 0.999$$

SECTION-3 (Maximum Marks: 24)

- This section contains SIX (06) questions. The answer to each question is a **NUMERICAL VLAUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, truncate round-off the value to TWO decimal places.
- Answer to each question will be evaluated according to the following making scheme:
 Full Marks : +4 If ONLY the correct numerical value is entered;
 Zero Marks : 0 In all other cases.

- *13. 5.00 mL of 0.10 M oxalic acid solution taken in a conical flask is titrated against NaOH from a burette using phenolphthalein indicator. The volume of NaOH required for the appearance of permanent faint pink color is tabulated below for five experiments. What is the concentration, in molarity, of the NaOH solution?

Exp. No.	Vol. of NaOH (mL))
1	12.5
2	10.5
3	9.0
4	9.0
5	9.0

Sol. 0.11

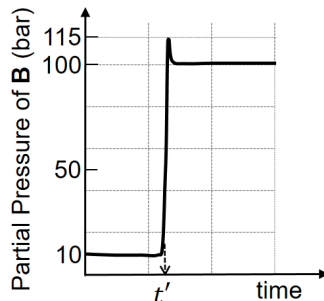
Vol. of NaOH used at equivalent point = 9 mL

So, mequivalent of NaOH used up to end point = $9 \times (M \times 1)$

$$5 \times 0.1 \times 2 = 9 \times M$$

$$M = \frac{1}{9} = 0.11$$

- *14. Consider the reaction $A \rightleftharpoons B$ at 1000 K. At time t' , the temperature of the system was increased to 2000 K and the system was allowed to reach equilibrium. Throughout this experiment the partial pressure of A was maintained at 1 bar. Given below is the plot of the partial pressure of B with time. What is the ratio of standard Gibbs energy of the reaction at 1000 K to that at 2000 K?



Sol. 0.25

$$K_1 \text{ at } 1000 \text{ K} = \frac{10}{1} = 10$$

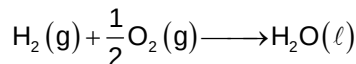
$$K_2 \text{ at } 2000 \text{ K} = \frac{100}{1} = 100$$

$$\Delta G_1^\circ = -RT_1 \ln 10 = -2.303RT_1$$

$$\Delta G_2^\circ = -2.303RT_2 \log 100 = -2 \times 2.303RT_2$$

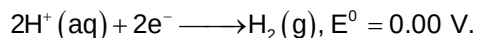
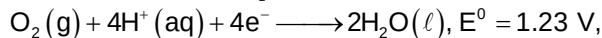
$$\frac{\Delta G_1^\circ}{\Delta G_2^\circ} = \frac{1 \times 1000}{2 \times 2000} = \frac{1}{4} = 0.25$$

15. Consider a 70% efficient hydrogen-oxygen fuel cell working under standard conditions at 1 bar and 298 K. Its cell reaction is



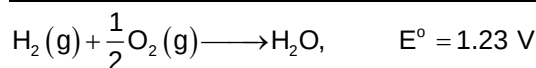
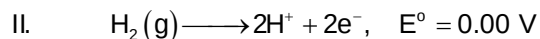
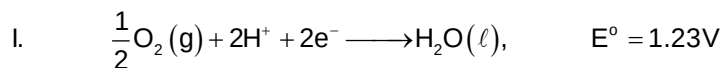
The work derived from the cell on the consumption of $10 \times 10^{-3} \text{ mol}$ of $\text{H}_2(\text{g})$ is used to compress 1.00 mol of a monoatomic ideal gas in a thermally insulated container. What is the change in the temperature (in K) of the ideal gas?

The standard reduction potentials for the two half – cells are given below.



Use $F = 96500 \text{ C mol}^{-1}$, $R = 8.314 \text{ J mol}^{-1}\text{K}^{-1}$.

Sol. 13.32



$$\Delta G^\circ = -2 \times 96500 \times 1.23 \times 1 \times 10^{-3} \times 0.7 = -166.173 \text{ J}$$

$$W = 166.173 \text{ J}$$

$$W_{\text{adiabatic}} = \frac{nR}{(\gamma - 1)}(T_2 - T_1)$$

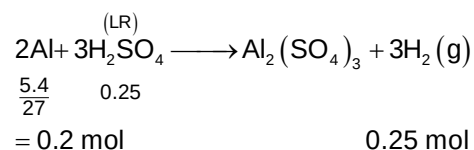
$$166.173 = \frac{1 \times 8.314}{\left(\frac{5}{3} - 1\right)} (T_2 - T_1)$$

$$\Delta T = \frac{166.173}{8.314} \times \frac{2}{3} = 13.32$$

- *16. Aluminium reacts with sulfuric acid to form aluminium sulfate and hydrogen. What is the volume of hydrogen gas in litre (L) produced at 300 K and 1.0 atm pressure, when 5.4 g of aluminium and 50.0 mL of 5.0 M sulfuric acid are combined for the reaction?

(Use molar mass of aluminium as 27.0 g mol^{-1} , $R = 0.082 \text{ atm L mol}^{-1} \text{ K}^{-1}$)

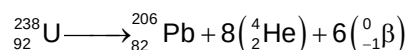
Sol. 06.15



$$V_{\text{H}_2} = \frac{0.25 \times 0.0821 \times 300}{1} = 6.1575 \text{ litre}$$

17. ${}^{238}_{92}\text{U}$ is known to undergo radioactive decay to form ${}^{206}_{82}\text{Pb}$ by emitting alpha and beta particles. A rock initially contained $68 \times 10^{-6} \text{ g}$ of ${}^{238}_{92}\text{U}$. If the number of alpha particles that it would emit during its radioactive decay of ${}^{238}_{92}\text{U}$ to ${}^{206}_{82}\text{Pb}$ in three half-lives is $Z \times 10^{18}$, then what is the value of Z?

Sol. 1.20



$$\text{Total moles of uranium} = \frac{68 \times 10^{-6}}{238}$$

$$= 0.286 \times 10^{-6} \text{ mol}$$

So, after 3 half-lives fraction left = 0.125

Fraction of uranium reacted = 0.875

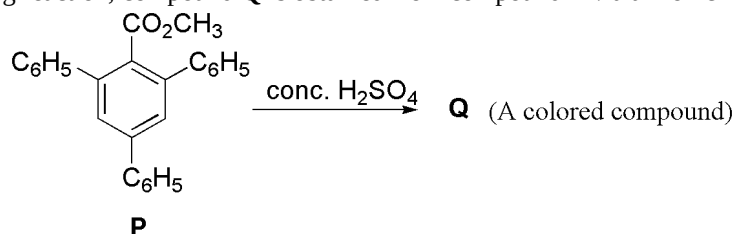
$$\text{So, number of moles of } \alpha \text{ - particles} = \frac{8 \times 68 \times 10^{-6}}{238} \times 0.875 \text{ mol}$$

$$\text{So, number of } \alpha \text{ - particle} = \frac{8 \times 68 \times 10^{-6} \times 0.875 \times 6.023 \times 10^{23}}{238}$$

$$= 12.046 \times 10^{17}$$

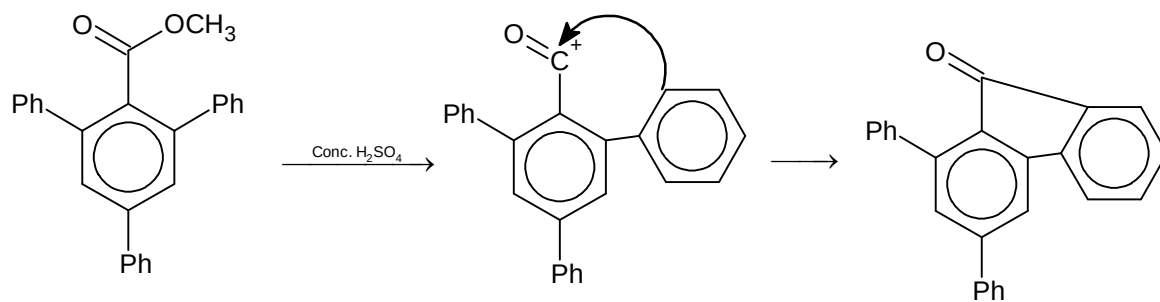
$$= 1.20 \times 10^{18}$$

18. In the following reaction, compound **Q** is obtained from compound **P** via an ionic intermediate.



What is the degree of unsaturation of **Q**?

Sol. 18



Degree of unsaturation = 18